Daily Geometry Drill #3

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Relative Positions · Radical Axis by subtraction · Common Chords · Orthogonal Circles · Tangency to Both Axes.

Section A Rapid Recognition

1. The circle $x^2 + y^2 - 12x - 16y + 75 = 0$ is tangent to the circle $x^2 + y^2 = 25$. Write down the number of common tangents of the two circles.

Ans: 3

2. The circles with centre $(0, 0)$ radius 5 and centre $(13, 0)$ radius 3 are disjoint. Write down the number of common tangents of the two circles.

Ans: 4

3. Write down the radical axis of the circles $x^2+y^2-6x+2y-11 = 0$ and $x^2+y^2-2x-4y+1 = 0$.

Ans: $2x - 3y + 6 = 0$

4. Write down whether the circles $x^2 + y^2 = 9$ and $x^2 + y^2 + 6x + 8y + 9 = 0$ are orthogonal (yes or no).

Ans: yes

5. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 10x + 5 = 0$ intersect in two points. Find the length of their common chord.

Ans: 8

6. Two circles have radii 3 and 5 and their centres are 4 units apart. Write down the number of common tangents.

Ans: 2

7. The circle $x^2 + y^2 - 10x - 10y + k = 0$ is tangent to both coordinate axes. Find k.

Ans: 25

8. The circle $x^2 + y^2 + 4x - 2y + C = 0$ is orthogonal to the circle $x^2 + y^2 - 4x + 4y - 1 = 0$. Find C.

Ans: −11

9. How many common tangents do the circles $x^2+y^2-4x-6y+12 = 0$ and $x^2+y^2+2x-10y+25 = 0$ have?

Ans: 4

10. Two circles have radii 7 and 3 and exactly 3 common tangents. Find the distance between their centres.

Ans: 10

Section B Manipulation Drills

1. How many common tangents do the circles $x^2+y^2-2x-4y-20=0$ and $x^2+y^2-6x-2y+6=0$ have?

A) 0

B) 1

C) 2

D) 3

E) 4

Ans: A) 0

2. The circle $x^2+y^2+ax+by+c=0$ passes through the origin, and its radical axis with the circle $x^2+y^2+6x-4y+10=0$ is the line $2x-y+5=0$. What is $a+b$?

A) 3

B) 5

C) -3

D) 0

Ans: D) 0

3. The circle $x^2 + y^2 - 4x + 2y - 5 = 0$ is orthogonal to $x^2 + y^2 + 6x - 2y + k = 0$. Find k.

A) -9

B) 9

C) -5

D) 14

Ans: A) -9

4. A circle is tangent to both coordinate axes in the first quadrant and passes through the point $(1, 8)$. Find the smaller possible radius.

Ans: 5

5. Two circles of equal radius 5 have centres $(0, 0)$ and $(6, 0)$. Their radical axis has equation?

A) $x = 3$

B) $x = 6$

C) $y = 3$

D) $y = 0$

Ans: A) $x = 3$

6. Two circles of radii 4 and 6 have centres 9 units apart. How many common tangents do they have?

A) 0

B) 1

C) 2

D) 3

E) 4

Ans: C) 2

7. A circle with centre $(7, 1)$ cuts the circle $x^2 + y^2 - 2x + 4y - 4 = 0$ at right angles. What is its radius?

A) $3\sqrt{5}$

B) $3\sqrt{5} - 3$

C) 3

D) 6

Ans: D) 6

8. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 12x - 6y + 25 = 0$ meet at P and Q. Find the midpoint of PQ.

Ans: $\left(\frac{10}{3}, \frac{5}{3}\right)$

9. A circle with its centre in the first quadrant touches both coordinate axes and the line $3x+4y=12$. What is the sum of the possible radii?

A) 1

B) 6

C) 7

D) 5

Ans: C) 7

10. The point $P=(1,t)$ lies on the radical axis $3x-4y+5=0$ of two circles. Find t.

Ans: 2

Section C Substitution & Structure

1. The circle $(x+2)^2+(y-1)^2=25$ and a circle with centre $(3,-2)$ meet at two points, and the line through those two points passes through the origin. What is the radius of the second circle? *(after TMUA 2021 Paper 1 Q1)*

A) $\sqrt{17}$

B) 5

C) $\sqrt{33}$

D) 3

Ans: C) $\sqrt{33}$

2. The circles $C_1 : (x-1)^2+y^2=4$ and $C_2 : (x-5)^2+(y-3)^2=9$ have radii 2 and 3. How many common tangents do C_1 and C_2 have?

A) 0

B) 1

C) 2

D) 3

E) 4

Ans: D) 3

3. The circles $x^2 + y^2 = 169$ and $x^2 + y^2 - 6x + 8y - 119 = 0$ meet at P and Q. What is the length PQ?

A) 12

B) 16

C) 24

D) 26

Ans: C) 24

4. A circle in the first quadrant is tangent to both coordinate axes and passes through the point $(9, 2)$. Its two possible radii are $r_1 < r_2$. What is $r_1 + r_2$?

A) 14

B) 17

C) 22

D) 34

Ans: C) 22

5. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 16x - 8y - 1 = 0$ meet at P and Q. What is the radius of the circle through P, Q and the point $(0, 1)$?

A) 9

B) $2\sqrt{13}$

C) $4\sqrt{5}$

D) 13

Ans: D) 13

6. A circle has equation $x^2 + ax + y^2 + by + c = 0$, where $a, b, c \neq 0$. Which condition is necessary and sufficient for the y-axis to cut a chord of length 2 from the circle? *(after TMUA 2021 Paper 2 Q15)*

A) $a^2 - 4c = 4$

B) $b^2 - 4c = 4$

C) $b^2 - 4c = 2$

D) $b^2 - 4c = 16$

Ans: B) $b^2 - 4c = 4$

7. The circles $x^2 + y^2 = 4$ and $(x-6)^2 + y^2 = 1$ have a common tangent with positive gradient that crosses the segment joining their centres. It makes an acute angle θ with the x-axis. What is $\tan\theta$? *(after TMUA 2020 Paper 1 Q16)*

A) $\dfrac{1}{2}$

B) $\dfrac{\sqrt{3}}{3}$

C) $\sqrt{3}$

D) $\dfrac{2\sqrt{5}}{5}$

Ans: B) $\frac{\sqrt{3}}{3}$

8. The three circles $C_1 : x^2+y^2+4x-1=0$, $C_2 : x^2+y^2-8y+11=0$, $C_3 : x^2+y^2+2x-12y+13=0$ have pairwise radical axes concurrent at R. What are the coordinates of R?

A) $(1, 1)$

B) $(1, -1)$

C) $(3, 1)$

D) $(1, 2)$

Ans: A) $(1, 1)$

Section D Challenge Ramp

1. The circles $x^2 + y^2 - 4x - 2y - 4 = 0$ and $x^2 + y^2 - 20x - 14y + 124 = 0$ have centres O_1 and O_2. A common external tangent touches them at P and Q respectively. What is the area of the quadrilateral O_1PQO_2?

A) 24

B) $16\sqrt{6}$

C) 40

D) $8\sqrt{6}$

Ans: B) $16\sqrt{6}$

2. The circle $(x+4)^2+(y+1)^2=64$ and a circle of radius 5 whose centre lies on the line $y=4$ have exactly one point in common. How many such circles of radius 5 are there? *(after TMUA 2019 Paper 1 Q6)*

A) 1

B) 2

C) 3

D) 4

Ans: B) 2

3. Three circles of radii 3, 4 and 5 are pairwise externally tangent. What is the area of the triangle formed by joining their centres?

A) $12\sqrt{5}$

B) 12

C) 84

D) 30

Ans: A) $12\sqrt{5}$

4. Two circles of radii 3 and 4 are orthogonal. What is the length of their common chord?

A) $\frac{12}{5}$

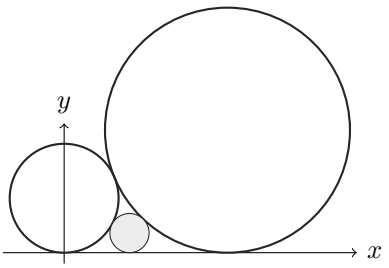
B) $\frac{24}{5}$

C) 6

D) $\frac{48}{5}$

Ans: B) $\frac{24}{5}$

5. The circles $x^2+y^2-8y=0$ and $x^2+y^2-24x-18y+144=0$ both touch the x-axis and touch each other. A third circle lies in the gap between them and the x-axis, and touches both circles and the x-axis. What is its radius? *(after Primer 1.4 P47)*



A) $\frac{6}{5}$

B) $\frac{36}{13}$

C) $\frac{169}{100}$

D) $\frac{36}{25}$

Ans: D) $\frac{36}{25}$